

Target validation paths — osteosarcoma-mets-dll3-h7r2

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

The case has two RNA-only targetable features, each with its own protein-level confirmation step. **DLL3 IHC (clone SP347)** gates ranks 2–3. **PRAME IHC plus HLA-A*02:01 typing** together gate ranks 4–5. The two pathways are independent: the patient may have neither, either, or both. All three workup tests can run in parallel on archival tissue and a single blood draw. If both workups return negative, the case has no within-scope recommendations, and the next conversation about standard 2L+ care is the treating team's, not Libby's.

DLL3 RNA expression

Before any DLL3-directed therapy decision: DLL3 IHC SP347 on archival FFPE, $\geq 1\%$ (preferably $\geq 25\%$) per NCT06788938's enrollment threshold. Turnaround is one to three weeks; cost is trivial relative to a treatment cycle. First, confirm SP347 assay availability at the treating institution — not every reference lab carries the Roche Tissue Diagnostics clone used in the tarlatamab development program.

Two refinements sit one tier down. Spatial heterogeneity is a known confounder in solid-tumor DLL3 (Zhang 2023): if multiple tumor blocks are available, IHC on a metastatic site as well as the primary refines confidence in whether the gating result generalizes. Neuroendocrine context (ASCL1 / NEUROD1 / chromogranin / synaptophysin / INSM1 panel) is research-grade for an osteosarcoma. DLL3 is normally a Notch-pathway target on neuroendocrine lineage, so an unexpectedly positive DLL3 IHC in a non-NEC tumor is worth contextualizing before the trial enrollment paperwork.

Germline TP53 sequencing (Li-Fraumeni panel) is a separate kind of finding. It does not affect tarlatamab eligibility, but late-teens / twenties osteosarcoma carries a meaningful prior probability for germline TP53. A positive result changes radiation-sensitivity planning, prompts screening for synchronous tumors, and triggers cascade testing for first-degree relatives. Discuss with a genetic counselor before ordering.

PRAME RNA expression

Two essential tests gate the entire PRAME-directed pathway: **PRAME IHC** confirms tumor protein expression, and **HLA-A*02:01 typing** confirms the patient can present the PRAME peptide. Every PRAME-directed ImmTAC and TCR-T in clinical development (IMA203, brenetafusp, IMC-P115C) is HLA-A02:01-restricted; *typing-negative patients are foreclosed even when PRAME IHC is strongly positive. PRAME IHC is positive in roughly 56% of osteosarcoma (Iura 2017, n=82); HLA-A02:01 prevalence is ~40–50% in Caucasian populations.* Order both alongside the DLL3 IHC: same block, same blood draw, same turnaround window.

Spatial heterogeneity refines confidence at the next tier. PRAME IHC on a metastatic site as well as the primary helps rule out the case where one biopsy hits a focal expression hotspot. Tumor NGS for B2M loss-of-function and HLA-A*02:01 loss-of-heterozygosity is a resistance check: TCR-mediated killing depends on intact antigen presentation, and tumor-specific HLA-LOH is a documented escape mechanism that the IHC and HLA-typing workup will miss by design. Quantifying baseline CD8 T-cell infiltrate density via IHC (or multiplex CD3 / CD8 / FoxP3) is the microenvironment companion: an immune-cold tumor predicts diminished response even when PRAME and HLA are both positive.

Where to order these assays

Assay	Provider	Contact
DLL3 IHC (clone SP347)	Foundation Medicine (FoundationOne CDx + IHC reflex)	test info · 1-888-988-3639
DLL3 IHC (clone SP347)	Mayo Clinic Laboratories	test info · 1-800-533-1710
DLL3 IHC (clone SP347)	NeoGenomics Laboratories	test info · 1-866-776-5907
DLL3 IHC (clone SP347)	Caris Life Sciences	test info · 1-888-979-8669
DLL3 IHC (clone SP347)	LabCorp / Esoterix Oncology	test info · 1-800-345-4363
PRAME IHC (clone EPR20330 or equivalent)	NeoGenomics Laboratories	test info · 1-866-776-5907
PRAME IHC	Caris Life Sciences	test info · 1-888-979-8669
PRAME IHC	Foundation Medicine	test info · 1-888-988-3639
PRAME IHC	LabCorp / Esoterix Oncology	test info · 1-800-345-4363
PRAME IHC	Mayo Clinic Laboratories	test info · 1-800-533-1710
HLA Class I high-resolution typing (HLA-A*02:01)	HistoGenetics	test info · 1-845-356-3801
HLA Class I typing	Versiti / Wisconsin Diagnostic Laboratories	test info · 1-800-245-3117
HLA Class I typing	ARUP Laboratories	test info · 1-800-242-2787
HLA Class I typing	Mayo Clinic Laboratories	test info · 1-800-533-1710
HLA Class I typing	Stanford Histocompatibility Laboratory	test info · 1-650-723-7960
Tumor NGS for HLA-LOH / B2M / antigen-presentation	Foundation Medicine (FoundationOne CDx)	test info · 1-888-988-3639
Tumor NGS (HLA-LOH / B2M)	Caris Life Sciences (Molecular Intelligence)	test info · 1-888-979-8669
Tumor NGS (HLA-LOH / B2M)	Tempus (xT)	test info · 1-800-739-4137
Tumor NGS (HLA-LOH / B2M)	Memorial Sloan Kettering (MSK-IMPACT)	test info · 1-212-639-2000
Tumor NGS (HLA-LOH / B2M)	NeoGenomics Laboratories (NeoTYPE Comprehensive)	test info · 1-866-776-5907
CD8 / TIL density (multiplex IHC)	NeoGenomics Laboratories (MultiOmyx)	test info · 1-866-776-5907
CD8 / TIL density	Caris Life Sciences	test info · 1-888-979-8669
CD8 / TIL density	Foundation Medicine	test info · 1-888-988-3639
Neuroendocrine IHC panel (ASCL1 / NEUROD1 / chromogranin / synaptophysin / INSM1)	ARUP Laboratories	test info · 1-800-242-2787
Neuroendocrine IHC panel	Mayo Clinic Laboratories	test info · 1-800-533-1710
Neuroendocrine IHC panel	LabCorp / Esoterix Oncology	test info · 1-800-345-4363
Neuroendocrine IHC panel	Quest Diagnostics	test info · 1-866-697-8378
Germline TP53 (Li-Fraumeni panel)	Invitae (Common Hereditary Cancers Panel)	test info · 1-800-436-3037
Germline TP53 panel	GeneDx	test info · 1-888-729-1206
Germline TP53 panel	Ambry Genetics (CancerNext)	test info · 1-866-262-7943
Germline TP53 panel	Myriad Genetics (MyRisk Hereditary Cancer)	test info · 1-800-469-7423

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.